Research Methodology

- Simply “how” a research is done.
- A systematic plan to perform a research on a specific topic/field.
- A detailed step by step approach to conduct the whole research.
- Describes the techniques and procedures used/applied in a research.
- Includes everything: planning, review, analysis, design, development, implement, experiment, analysis, result, findings, outcome, contribution, significance.
- Explains what method/theory/algorithm/technique is chosen and why, what data is needed, how to do experiment/analysis,…….
- Includes a research’s overall validity and reliability.

Research Methodology

- Research methodology differs from research methods.
  - ➢Methodology is the general approach of conducting a research.
  - ➢Method focuses on the specific tools/techniques/strategies a researcher uses/follows to design/develop/analyze the work.

Methodology vs. Methods

Research Method	Research Methodology
Refers to the specific techniques or procedures used to collect and analyze data within a particular research project.	Refers to the overall theoretical framework or approach that guides the research process.
Examples include surveys, interviews, experiments, case studies, and observational studies.	Examples include quantitative, qualitative, mixed methods, action research, and participatory research.
Focuses on the practical aspects of data collection and analysis.	Focuses on the theoretical and philosophical foundations of research.
It involves deciding which methods to use based on the research question and the available resources.	It involves making decisions about the overall approach and strategy for conducting research.
Often involves following a standardized set of steps or procedures.	Often involves a more flexible and adaptable approach.
It can be used within different research methodologies.	Determines the overall structure and direction of the research project.

Types of Research/Research Method

- There are many ways to categorize research based on the discipline and field.
- It depends on the types of knowledge, types of data a researcher aims to produce/analyze.
- In general, the basic two types of research are:
  - ➢ Fundamental or Basic research
  - ➢ Applied research

Fundamental or Basic Research

- Aims to develop/expand knowledge, theories, principles, and predictions.
- Not concerned with the solving of a practical problem with immediate interest.
- May not lead to immediate use or application.
- Provides foundation for applied research.
- Seeks generalization.
- Seeks answers to What, Why, or How in the research area.
- Zhang, Jian & Zhang, Wenhui & Zhan, Naijun & Shen, Yidong & Chen, Haiming & Zhang, Yunquan & Wang, Yongji & Wu, Enhua & Wang, Hongan & Zhu, Xue-Yang. (2008). Basic research in computer science and software engineering at SKLCS. Frontiers of Computer Science in China. 2. 1-11. 10.1007/s11704-008-0001-3.

Applied research

- Aims to develop techniques, products and procedures.
- Solves problems using well known/established accepted theories and principles.
- Has immediate application, solution or use.
- Helpful for basic research.
- Experiment, case studies, interdisciplinary research are applied research.
- Hassani, Hossein. Research Methods in Computer Science: The Challenges and Issues. *ArXiv* abs/1703.04080 (2017).

Further Types of Research

Basic or Applied research can be:

- Qualitative research
  - Non-numerical and descriptive.
  - Focuses on words and meaning.
  - Exploratory.
- Quantitative research
  - Numerical and non-descriptive.
  - Applies statistics and mathematics.
  - Focuses on number and statistics.
  - Conclusive.
- Mixed research
  - Mixing of both qualitative and quantitative research.

Example???

More types of Research

- Exploratory research
  - ➢ It’s used to gain insights, generate ideas, and clarify concepts, i.e., a first step before conducting more structured research.
  - ➢ Includes literature search, focus group interview etc..
  - ➢ Uses interviews, observations, or open-ended surveys.
  - ➢ Identifies key issues and key variables.
  - ➢ Provides better understanding.
  - ➢ Tests feasibility of the extensive study.
  - ➢ Determines the best methods for the subsequent study.
  - ➢ Basically seeks the answer of “What is going on?”.
  - ➢ Example: ??

- Descriptive research
  - ➢ Explains the problems, situations, behavior, characteristics of the phenomenon being studied.
  - ➢ Basically seeks the answer of “What is this?”.
  - ➢ Beneficial for other research.
  - ➢ Example: case studies, census data analysis.

More types of Research

- Explanatory research
  - ➢Finds relationship between variables; events, situations, dimensions etc..
  - ➢To understand how one variable influences another.
  - ➢Explains reasons.
  - ➢Basically seeks answers to Why and How.
  - ➢Experiments, statistical analysis, …
- Comparative research
  - ➢Compares between two units.
  - ➢Finds similarities and differences between two or more subjects such as systems, methods, tools, groups, or technologies.
  - ➢Identifies “Which one performs better”, How do they differ in efficiency?”, “Tradeoff(s) between them?”…….

And Many More….

Scientific vs. Social Science Research

- Social Science research
  - Investigates/analyzes the behavior/nature of the human in social life.
  - May be qualitative, quantitative or both.
  - Researcher’s personal feelings might get involved in findings; subjective or biased.
  - Takes a sample of the population for research.
  - Difficult to repeat.
  - Takes place in the society.
- Scientific research
  - Concerned with natural science.
  - Generates new knowledge in this area.
  - Investigates through empirical and measurable techniques.
  - Scientific research is objective.
  - Repetitive through the change of values of variables.
  - Takes place in the laboratory.

Constructive Research

- Widely used in Software Engineering and Computer Science.
- Involves the development/creation/building of software, application, model, diagram, plan, system, framework, etc.
- Solves problems and improves/modifies/updates existing system/performance.
- Engineering research tradition.
- Involves other approaches.

Literature Review

- **Review**: a formal assessment of something with the intention of instituting change if necessary**.**
- A literature review is a comprehensive summary of previous research on a topic.
- A critical summary and evaluation of existing research on a specific topic.
- The literature review surveys scholarly articles, books, and other sources relevant to a particular area of research.
- The review enumerates, describes, summarizes, objectively evaluates and clarifies the previous research.
- The literature review acknowledges the work of previous researchers.

Literature Review

- Purpose:
  - ➢ To find the research gap/limitation to be addressed.
  - ➢ It gives a theoretical base for the future research.
  - ➢ To avoid duplication.
  - ➢ To support the methodology/theoretical framework.
  - ➢ To focus on technical depth, not paper count.
  - ➢ In CSE, generally, algorithmic, architectural, or system design choices are analyzed.
  - ➢ To identify what can be improved or extended.
  - ➢ Helps to select/choice datasets, models, architectures, evaluation metrics, etc.

Literature Review

- Available sources:
  - ➢EEE Xplore
  - ➢ACM Digital Library
  - ➢Google Scholar
  - ➢SpringerLink
  - ➢ScienceDirect
  - ➢Scopus

Literature Review

How to conduct/Steps of literature review

- ➢ Search for the relevant topic: make list of keywords, find the source.
- ➢ Evaluate and select the source: take notes and cite the sources.
- ➢ Take short notes on the objective, research questions, methodology, performance metrics, findings, and future plan/limitation(s).
- ➢ Prepare a summery of the reviewed papers.
- ➢ Identify the key issues of the reviewed papers.
- ➢ Make a plan for future work.

- Literature review ends with a summary of findings, identifying gaps, and plan for next steps.

Literature Review

- Common mistakes:
  - ➢Just summarizing sources without any analysis.
  - ➢Not organizing information properly.
  - ➢Using outdated or irrelevant sources.
  - ➢Failing to cite properly.

Reference List

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- https://www.scribbr.com/methodology/types-of-research/
- https://www.iedunote.com/basic-research